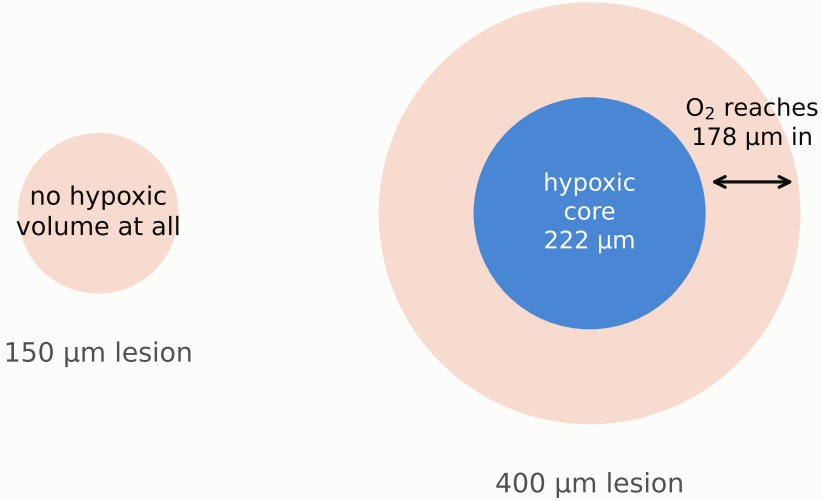
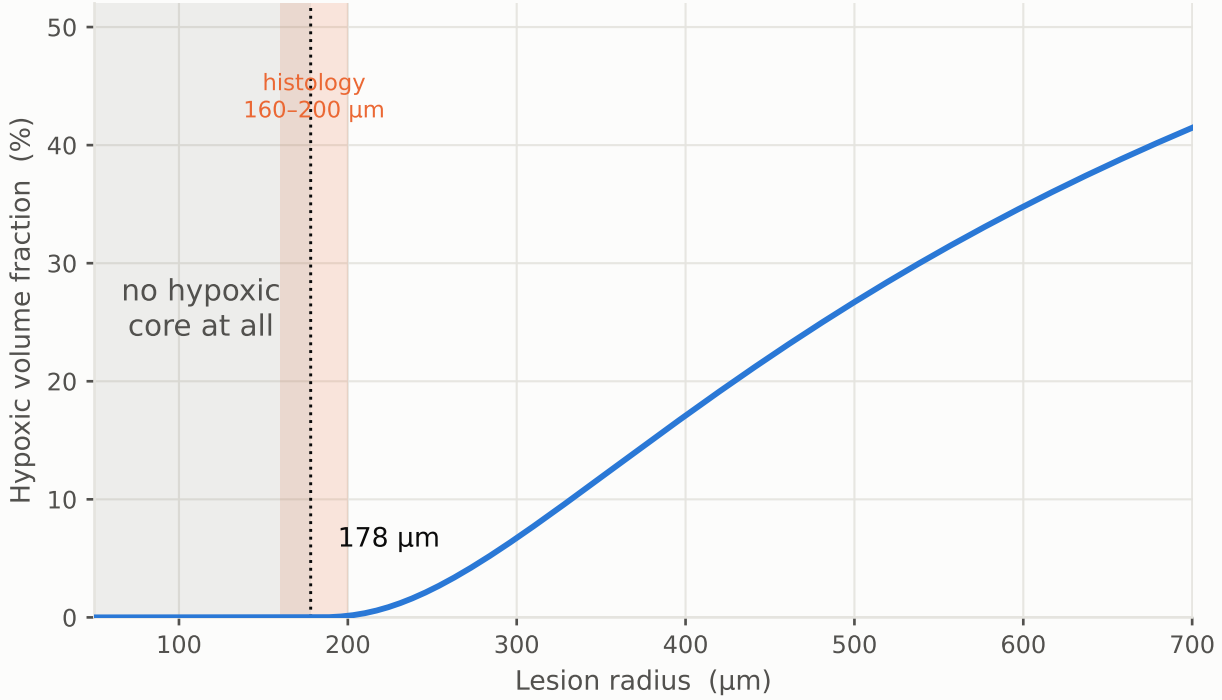
Oxygen reaches a fixed depth in from the surface



A safety floor that needs no calibration



Pure geometry: the hypoxic core is max(R - d, 0), so below the penetration depth there is no hypoxic volume to sense. Distant micro-disease needs another route.